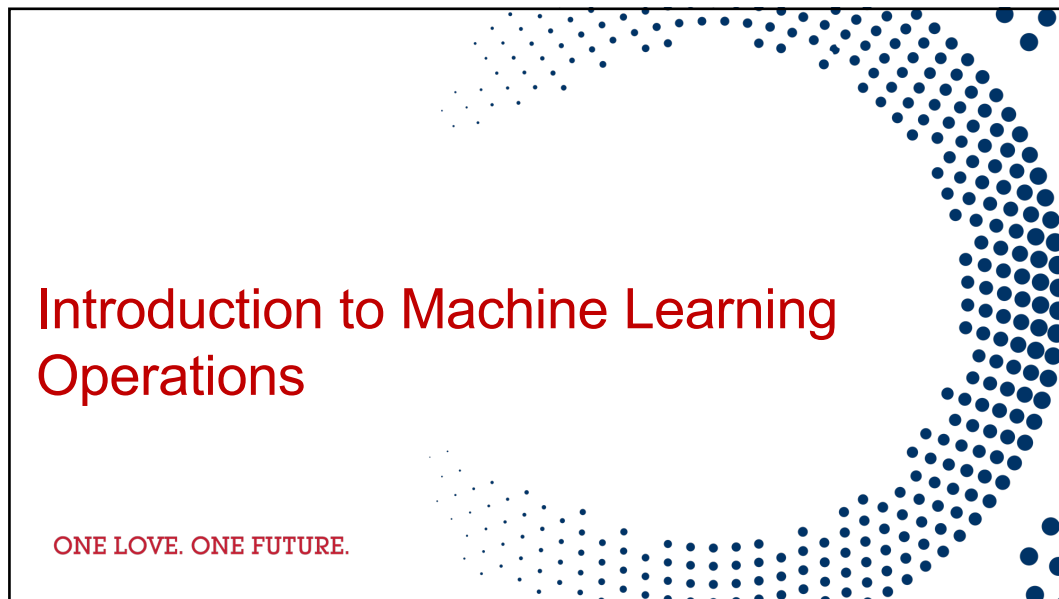




1

About this course	
Name	Vận hành hệ thống học máy (Machine Learning Operations)
Course ID	IT5414
Credit	2
Slides, group registration, and announcement	– On Microsoft Teams qw3hz1n
Time & location	

MLOps is a combination of data science and DevOps principles that speeds up model deployment and ensures product quality. This course provides an overview of the MLOps process, including planning, developing, testing, deploying, managing, and operating machine learning models in a production environment


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Evaluation

- Project: (including coding, report, representation)
 - Teamwork : 2-3 students/group
 - Topics : to be defined
- Final examination

3

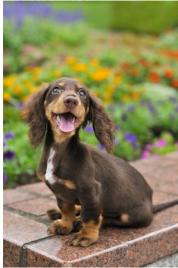
ML research vs. ML production

	Research	Production
Objectives	Model performance*	Different stakeholders have different objectives

4

Stakeholder objectives

ML team
highest



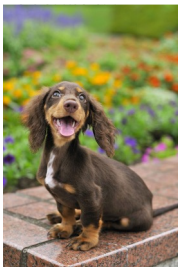
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Stakeholder objectives

ML team
highest



Sales
sells more



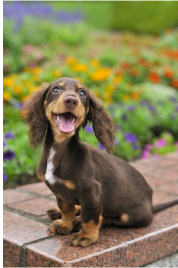
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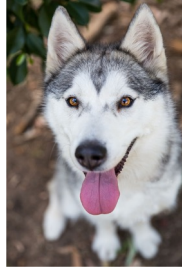
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Stakeholder objectives

ML team
highest



Sales
sells more



Product
fastest



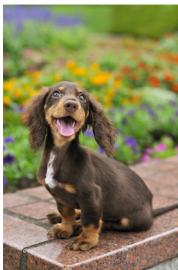
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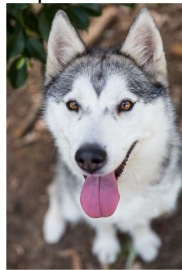
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Stakeholder objectives

ML team
highest



Sales
sells more



Product
fastest



Manager
maximizes profit



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Leaderboard-style ML

- More comprehensive utility function
 - Model performance (e.g. accuracy)
 - Latency
 - Prediction cost
 - Interpretability
 - Robustness
 - Ease of use (e.g. OSS tools, community support)
 - Hardware requirements
- Adaptive to different use cases
 - Instead of a leaderboard for each dataset/task, the leaderboard adapts to each company's needs
- Dynamic datasets
 - Realistic distribution shifts with different types of shifts



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Computational priority

	Research	Production
Objectives	Model performance	Different stakeholders have different objectives
Computational priority	Fast training, high throughput	Fast inference , low latency

generating predictions



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Latency matters

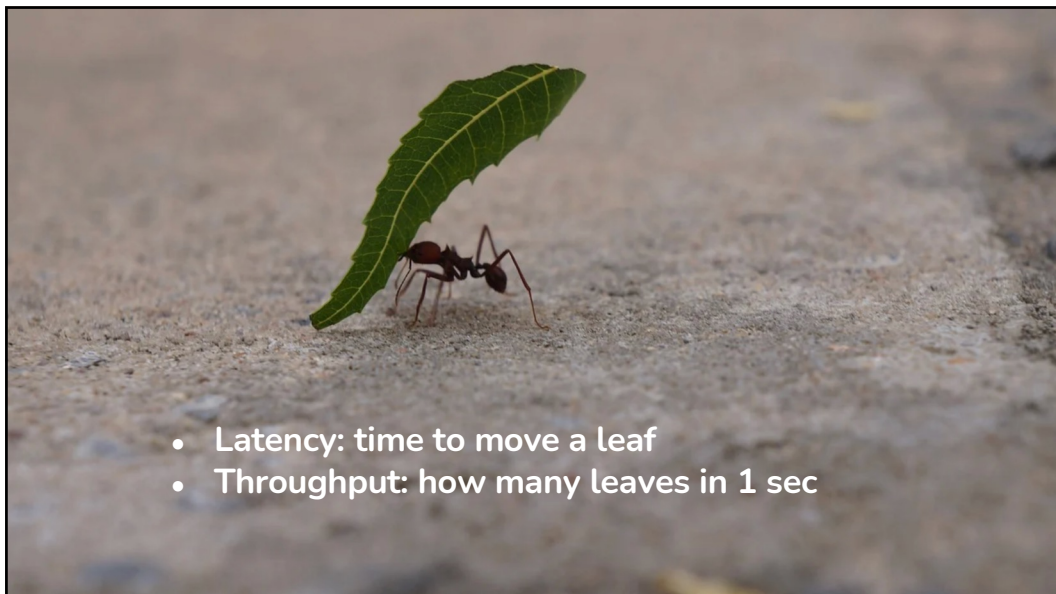
- 100ms delay can hurt conversion rates by 7% ([Akamai study](#) '17)
- 30% increase in latency costs 0.5% conversion rate ([Booking.com](#) '19)
- 53% phone users will leave a page that takes >3s to load ([Google](#) '16)



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- Latency: time to move a leaf
- Throughput: how many leaves in 1 sec

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ML in research vs. in production

	Research	Production
Objectives	Model performance	Different stakeholders have different objectives
Computational priority	Fast training, high throughput	Fast inference, low latency
Data	Static	Constantly shifting

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Data

Research	Production
• Clean • Static • Mostly historical data	• Messy • Constantly shifting • Historical + streaming data • Biased, and you don't know how biased • Privacy + regulatory concerns

THE COGNITIVE CODER

By [Armand Ruiz](#), Contributor, InfoWorld | SEP 26, 2017 7:22 AM PDT

The 80/20 data science dilemma

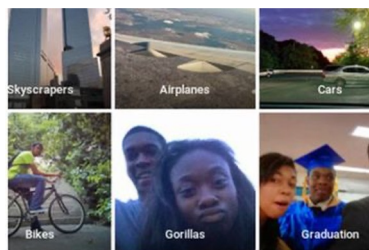
Most data scientists spend only 20 percent of their time on actual data analysis and 80 percent of their time finding, cleaning, and reorganizing huge amounts of data, which is an inefficient data strategy

ML in research vs. in production

	Research	Production
Objectives	Model performance	Different stakeholders have different objectives
Computational priority	Fast training, high throughput	Fast inference, low latency
Data	Static	Constantly shifting
Fairness	Good to have (sadly)	Important

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Fairness



Google Shows Men Ads for Better Jobs

by Krista Bradford | Last updated Dec 1, 2019



The Berkeley study found that both face-to-face and online lenders rejected a total of 1.3 million creditworthy black and Latino applicants between 2008 and 2015. Researchers said they believe the applicants "would have been accepted had the applicant not been in these minority groups." That's because when they used the income and credit scores of the rejected applications but deleted the race identifiers, the mortgage application was accepted.

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ML in research vs. in production

	Research	Production
Objectives	Model performance	Different stakeholders have different objectives
Computational priority	Fast training, high throughput	Fast inference, low latency
Data	Static	Constantly shifting
Fairness	Good to have (sadly)	Important
Interpretability*	Good to have	Important



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Interpretability



Zoom poll: which one would you want as your surgeon?

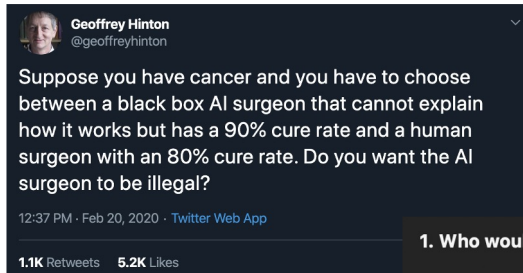


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Interpretability



Result from the Zoom poll
last year

1. Who would you rather pick?

AI Surgeon (90% accuracy) (44) 67%

Human Surgeon (80% accuracy) (22) 33%

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ML in research vs. in production

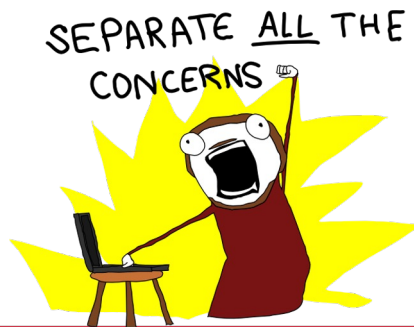
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Fairness	Good to have (sadly)	Important
Interpretability	Good to have	Important

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Traditional software

Separation of Concerns is a design principle for separating a computer program into distinct sections such that each section addresses a separate concern

- Code and data are separate
 - Inputs into the system shouldn't change the underlying code



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Image by [Arda Cetinkaya](#)

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ML systems

- Code and data are tightly coupled
 - ML systems are part code, part data
- Not only test and version code, need to test and version data too

the hard
part

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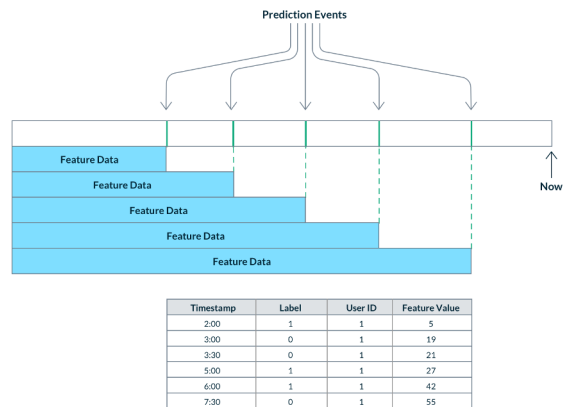
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Test and version data

- Extremely hard to ensure correctness in time

Don't panic.
We'll revisit
later!



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[Back to the Future: Solving the time-travel problem in machine learning - Tecton](#)

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ML systems: version data

- Line-by-line diffs like Git doesn't work with datasets
- Can't naively create multiple copies of large datasets
- How to merge changes?



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How to ...

- Validate data correctness?
- Test features' usefulness?
- Detect when the underlying data distribution has changed?
- Know if the changes are bad for models without ground truth labels?
- Detect malicious data?
 - Not all data points are equal (e.g. scans of cancerous lungs are more valuable)
 - Bad data might harm your model and/or make it susceptible to attacks

ML systems: data poisoning attacks

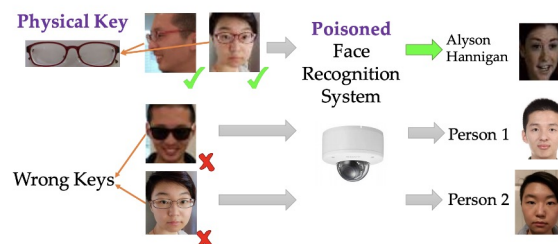


Fig. 1: An illustrating example of backdoor attacks. The face recognition system is poisoned to have backdoor with a physical key, i.e., a pair of commodity reading glasses. Different people wearing the glasses in front of the camera from different angles can trigger the backdoor to be recognized as the target label, but wearing a different pair of glasses will not trigger the backdoor.



REMEMBER WHEN A BILLION-PARAMETER MODEL WAS CONSIDERED LARGE

PEPPERIDGE FARM REMEMBERS

SWITCH TRANSFORMERS: SCALING TO TRILLION PARAMETER MODELS WITH SIMPLE AND EFFICIENT SPARSITY

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William Fedus*
 Google Brain
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Barret Zoph*
 Google Brain
 barretzoph@google.com

Noam Shazeer
 Google Brain
 noam@google.com

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Engineering challenges with large ML models

- Too big to fit on-device
- Consume too much energy to work on-device
- Too slow to be useful
 - Autocompletion is useless if it takes longer to make a prediction than to type
- If unit/CI tests take hours, the development cycles will stagnate

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5. ML production myths



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Myth #1: Deploying is hard



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Myth #1: Deploying is hard

Deploying is easy. Deploying reliably is hard



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Myth #2: You only deploy one or two ML models at a time



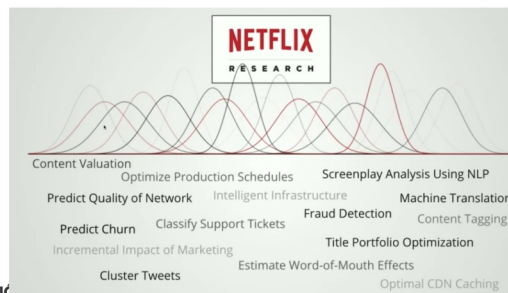
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Myth #2: You only deploy one or two ML models at a time

Booking.com: 150+ models, Uber: thousands



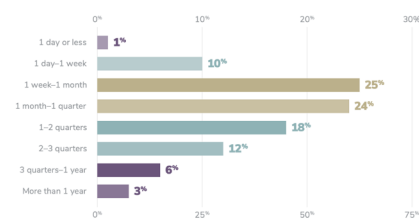
Myth #3: You won't need to update your models as much

DevOps: Pace of software delivery is accelerating

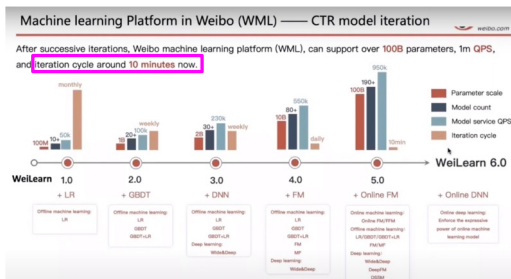
- Elite performers deploy **973x** more frequently with **6570x** faster lead time to deploy ([Google DevOps Report, 2021](#))
- DevOps standard (2015)
 - Etsy deployed 50 times/day
 - Netflix 1000s times/day
 - AWS every 11.7 seconds

DevOps to MLOps: Slow vs. Fast

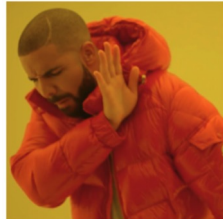
Only 11% of organizations can put a model into production within a week, and 64% take a month or longer



We'll learn how to do minute-iteration cycle!



Accelerating ML Delivery



How
often **SHOULD**
I update
my models?



How often
CAN I update
my models?



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ML + DevOps = 



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Myth #4: ML can magically transform your business overnight



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Myth #4: ML can magically transform your business overnight

Magically: possible
Overnight: no



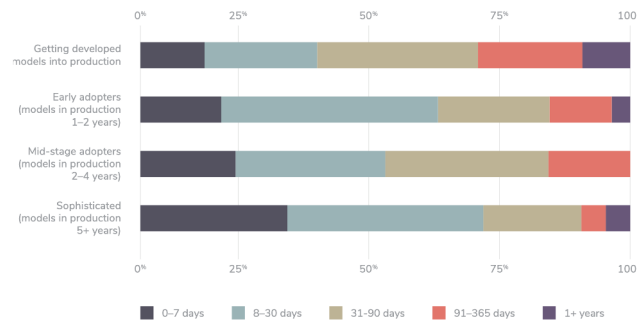
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Efficiency improves with maturity

Model deployment timeline and ML maturity



[2020 state of enterprise machine learning](#) (Algorithmia, 2020)

ML engineering is more engineering than ML

MLEs might spend most of their time:

- wrangling data
 - understanding data
 - setting up infrastructure
 - deploying models
- instead of training ML models



Myth #5: Most ML engineers don't need to worry about scale



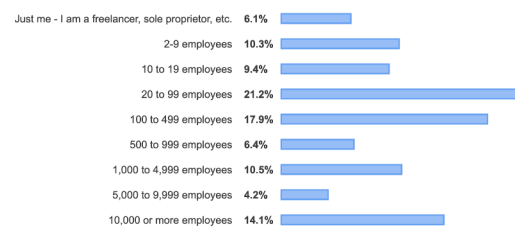
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Myth #5: Most ML engineers don't need to worry about scale

Company Size



71,791 responses

[StackOverflow Developer Survey 2019](#)



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What this course is not

- How different machine learning models works

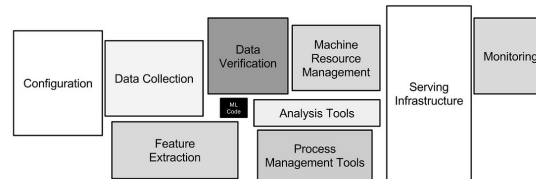


Figure 1: Only a small fraction of real-world ML systems is composed of the ML code, as shown by the small black box in the middle. The required surrounding infrastructure is vast and complex.



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Let's start where it all began

Machine learning in production is fantastic
BUT

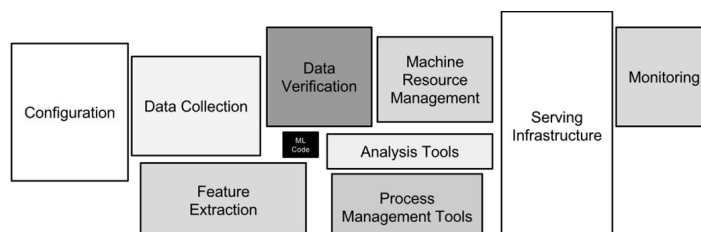


Massive technical depth is incurred if not careful

Hidden Technical Debt in Machine Learning Systems

D. Sculley, Gary Holt, Daniel Golovin, Eugene Davydov, Todd Phillips
{dsculley, gholt, dgg, edavydov, toddphillips}@google.com
Google, Inc.

Dietmar Ebner, Vinay Chaudhary, Michael Young, Jean-François Crespo, Dan Dennison
{ebner, vchaudhary, mxyoung, jfcrespo, dennison}@google.com
Google, Inc.

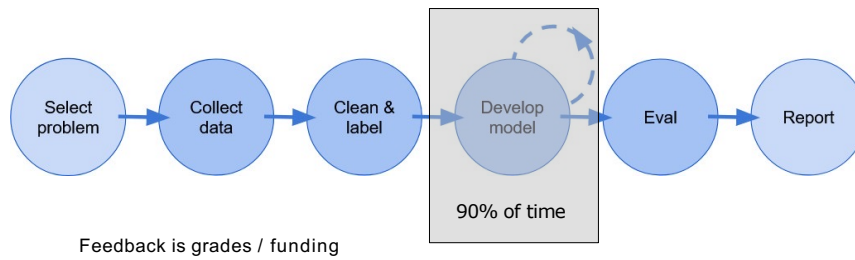


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Why do we focus on modelling?

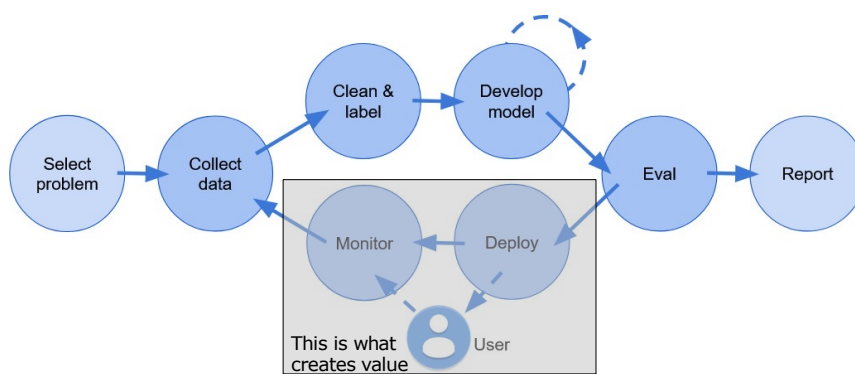


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Machine Learning in the real world



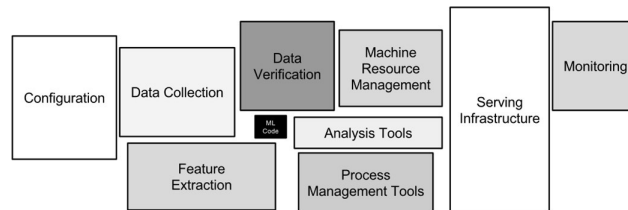
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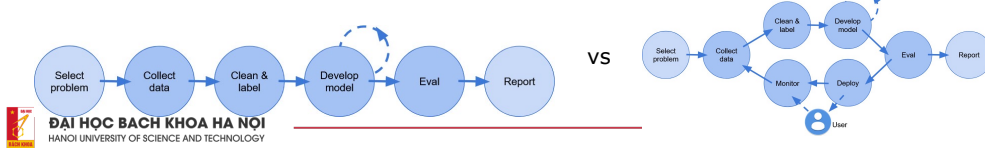
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Key observations

1. Machine Learning in production is much more than doing ML modelling



2. Machine Learning in production is a cycle



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The other stuff is DevOps

DevOps = Developer operations

- Dates to late 80s and early 90s
- Around 2007/2008 rose to popularity to remove the separation between software development with its operations part/IT department

💡 This is both a joke and not.

💡 MLOps is directly derived from DevOps.

💡 Therefore, let's try to understand DevOps first.



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So, what is DevOps?

This is the closest to a definition that I could find:

DevOps is a set of practices that combines software development (*Dev*) and IT operations (*Ops*). It aims to shorten the systems development life cycle and provide continuous delivery with high software quality. It's an combination of human mindset, processes and technologies that continuously creates value.



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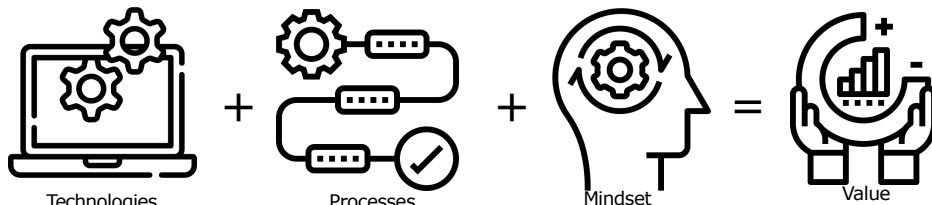
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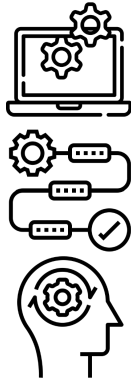


Technologies
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Technology, Processes, Mindset



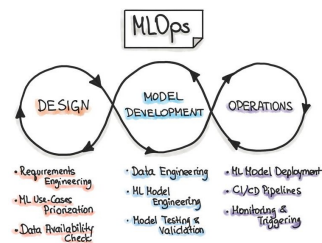
Use technologies that support the different parts of the lifecycle

Implement processes to make sure everyone is in sync about the lifecycle

Always consider all part of the lifecycle, not just its parts

But then MLOps must be...

Is a set of **tools**, **processes**, and **mindset** that aim to make **ML Lifecycle** reproducible, trackable, testable and maintainable to continuously create value.

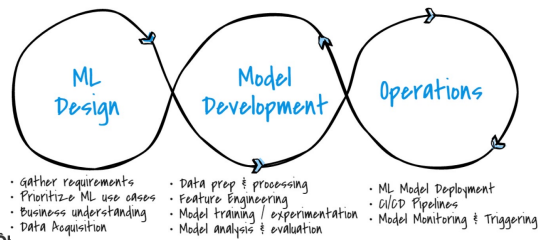


Let's look at the different phases of MLOps

Data phase

- 🔥 Business understanding
- 🔥 Data understanding
- 🔥 Designing the ML-powered software

Machine Learning Operations (MLOps)



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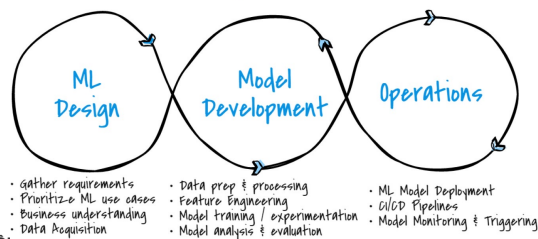
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Model phase

- 🔥 Model engineering
- 🔥 Data engineering
- 🔥 Deliver a stable quality ML model that we will run in production

Machine Learning Operations (MLOps)



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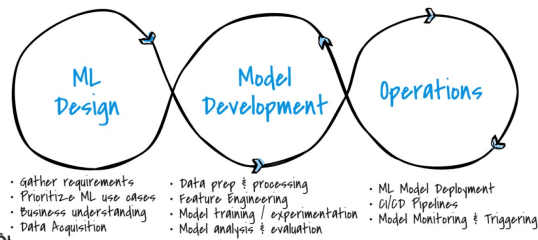
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Operations phase

- 🔥 Deliver the previously developed ML model in production
- 🔥 Testing, versioning, continuous delivery, and monitoring

Machine Learning Operations (MLOps)



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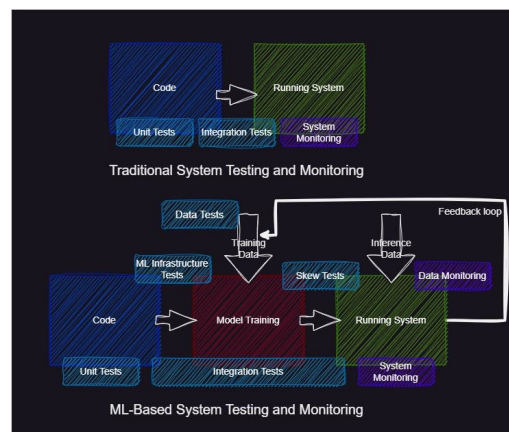
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If DevOps exist, then why do we need MLOps?



Because data changes everything



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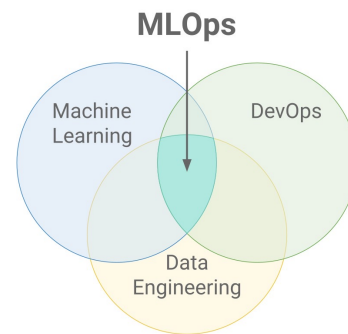
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What makes an MLOps engineer?

A mix of

- Software developing
- Machine Learning
- Data engineering

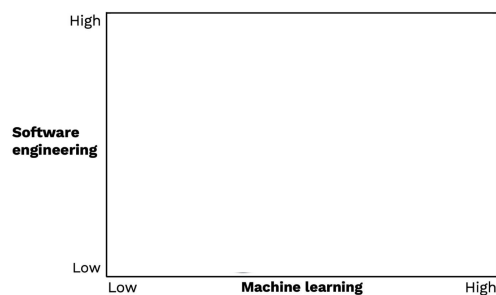


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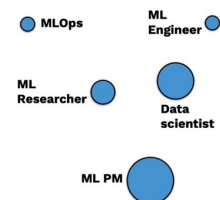
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Where's waldo?



Size of bubble =
communication /
technical writing

Where should the different
positions be?



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According to stable diffusion

<https://stablediffusionweb.com/>

Prompt:

"Machine Learning operations engineer"

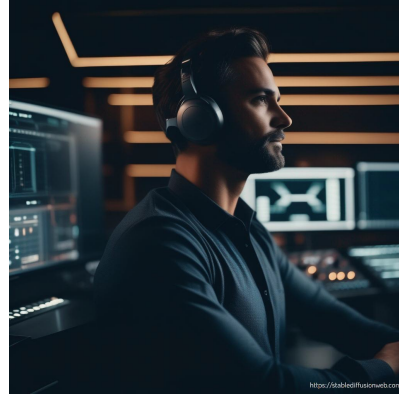
A MLOps engineer is

🔥 Buff

🔥 Locked in

🔥 Many screens

Its not completely wrong



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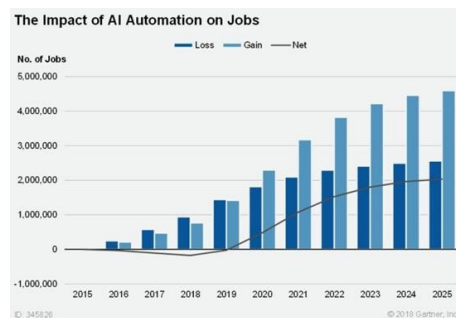
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Why does companies care about MLOps

Having automated model deployed with errors can cost A LOT of money:

"A famous example of the dangers here was Knight Capital's system losing \$465 millions in 45 minutes, apparently because of unexpected behavior from obsolete experimental codepaths"

– Hidden Technical depth in Machine Learning Systems



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Why is MLOps hot?

1. Growing Adoption of Machine Learning (ML)
2. Complexity of ML Workflow
3. Bridge between Development and Operations
4. Need for Collaboration
5. Ensuring Model Governance and Compliance
6. Automation and Scalability
7. Continuous Integration and Continuous Deployment (CI/CD)
8. Infrastructure Orchestration
9. Adoption of Cloud Services
10. Business Impact



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Open AI study

🔍 What is the contributes to the success of OpenAI?

- 💡 Funding
- 💡 Data
- 💡 People
- 💡 Compute
- 💡 Service contract with Microsoft

Microsoft fired 10,000 workers in the same breath as the invested \$10B in OpenAI



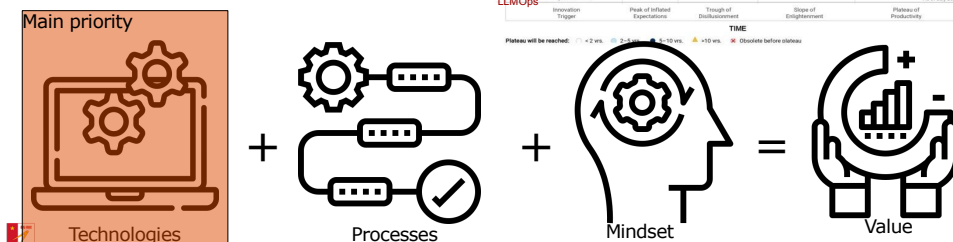
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Trends in MLOps

MLOps has been trending for a couple of
Tools have been the main priority



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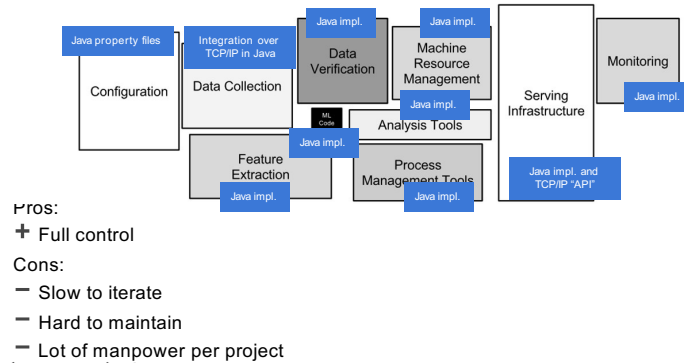
Choosing the right tool for the job

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Looking back

MLOps around 2006 = write everything from scratch



Today we have options

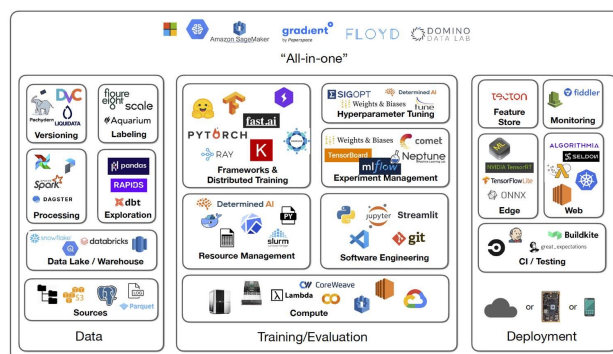
- There is a tool for everything you need

Pros:

- + Easy to get started
- + Easy to iterate

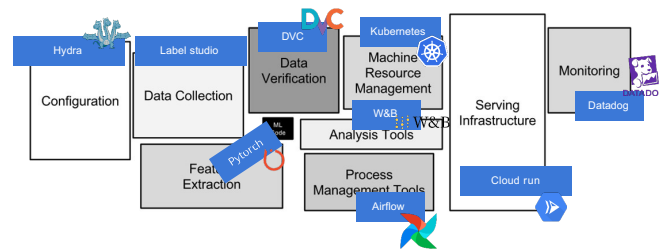
Cons:

- Framework integration can be really hard
- Hard to compare frameworks



MLOps now

Pick a *stack* of tools



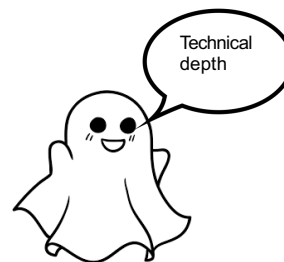
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Not picking the right tool leads to TD

- In a nutshell MLOps is about dealing reducing technical debt
 - ⚠️ Technical debt is the implied cost of future reworking required when choosing an easy but limited solution instead of a better approach that could take more time



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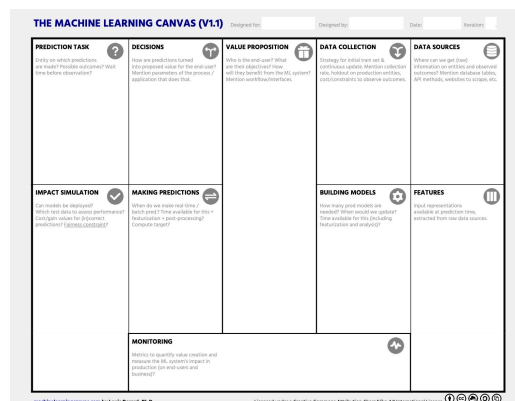
MLOps is full stack

- In MLOps we embrace the full stack of problems that comes from the full lifecycle. Especially integration problems.
- Criteria for what goes into the stack (4Cs):
 - 💡 Cost
 - 💡 Coverage
 - 💡 Complexity
 - 💡 Community
- Whenever we need to pick one tool over the other, we need to consider these 4 criteria.
- And most time this is not possible without actually trying to use both.



MLOps at its core is...

- ...delivering value for business 🏆
- ...thinking about the hole pipeline, not just data and model 🤖
- ...accounting for long term goals from the start 📅



Meme of the day



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